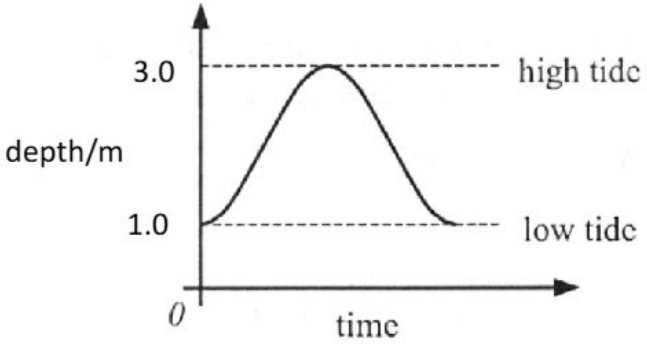
12 The water depth in a harbour varies in a simple harmonic manner between 1.0 m at low tide and 3.0 m at high tide, as shown in the figure below. The duration between successive low tides is 12 hours.



A certain boat requires a minimum depth of 1.5 m to enter and to exit the harbour.

If the boat enters the harbour at a depth of 1.5 m as the tide was rising, what is the maximum duration that it can stay in the harbour if it wants to exit the harbour within this tide cycle?

A 1.0 h **B** 2.0 h **C** 8.0 h **D** 10 h